TD n° 1: Intégrales doubles et triples

Exercice 1 Calculer les intégrales doubles suivantes :

a). $\int\int_D (2x-y)^2 dxdy$; $D = \{$ le parallélogramme limité par les droites d'équation $y = x,\ y = 2x,\ y = x+1,\ y = 2x-2\}$

b). $\int\int_D xy^2 dxdy$; $D = \{(x,y) \in \mathbb{R}^2;\ x+y \geqslant 0,\ x^2+y^2 \leqslant 1\}$

c). $\int\int_D (x+2y)^2 dxdy$; $D = \{$intérieur du triangle de sommets $A(0,0), B(1,1), C(2,-1)\}$

d). $\int\int_D \frac{xy}{(x^2+y^2)^2} dxdy$; $D = \{$l'ensemble des points du disque de centre 0 et de rayon 1 tq $x+y \geqslant 1\}$

Exercice 2 Calculer $I = \int\int_D f(x,y)dxdy$ en utilisant le changement de variables indiqué :

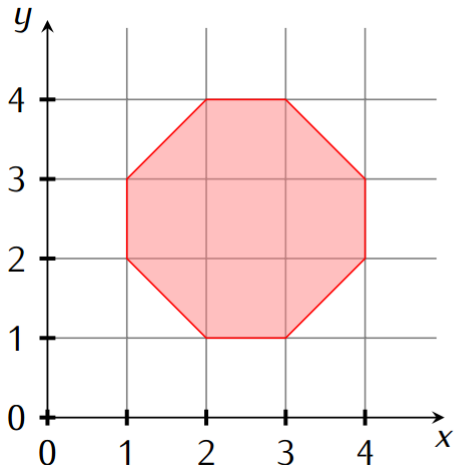
a). $f(x,y) = (x-y)^2$; $D = \{$l'ensemble des points du disque de centre O et de rayon 1, tels que$0 \leqslant y \leqslant x\}$; coordonnées polaires.

b). $f(x,y) = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^{-1/2}$; $D = \left\{(x,y) \in \mathbb{R}^2;\ \frac{x^2}{a^2} + \frac{y^2}{b^2} \leqslant 1\right\}$; $x = a\rho cos\theta$; $y = b\rho sin\theta$; $\theta \in [0, 2\pi]$; $\rho \in [0,1]$.

c). $f(x,y) = 1$; $D = \{$le domaine limité par les courbes d'equations $y = ax$; $y = x/a$; $y = b/x$; $y = 1/bx$; $(a > 1; b > 1; x > 0)$ changements de variables $x = u/v$; $y = uv\}$

Exercice 3 Soit D la plaque homogène dans la figure ci-dessous. Sans faire de calculs d'integrales, en déduire :

$$\int\int_D dxdy; \int\int_D xdxdy; \quad \int\int_D ydxdy.$$



Exercice 4 Après avoir représenté graphiquement la région :

$$D = \{(x,y) \in \mathbb{R}^2;\ y \leqslant 0,\ 1 \leqslant x^2+y^2 \leqslant 4\}$$

Calculer les coordonnées de son centre de gravité en supposant la région D homogène.

Tournez la page

Exercice 5 Calculer les intégrales triples suivantes :

a). $\int\int\int_D (x+y+z)^2 dxdydz$; où D est le domaine limité par les plans d'équation $x=0,\ y=0,\ z=0,\ x+y+z=1$

b). $\int\int\int_D (x-y+z)^2 dxdydz$; où D est le domaine limité par les plans d'équation $x=0,\ y=0,\ z=0,\ x+z=1,\ y+z=1$

c). $\int\int\int_D \frac{z}{\sqrt{x^2+y^2}} dxdydz;\quad D=\left\{(x,y,z)\in\mathbb{R}^3;\quad x^2+y^2<a^2 \text{ et } 0<z<a\right\}$ avec $a\in\mathbb{R}_+^*$.

Exercice 6 En utilisant les coordonnées cylindriques, calculer :

$$\int\int\int_D \frac{z}{\sqrt{x^2+y^2}} dxdydz;\quad D=\left\{(x,y,z)\in\mathbb{R}^3;\ 0<z\leqslant b\ \ 0<x^2+y^2<2ax;\ \ a>0\right\}$$

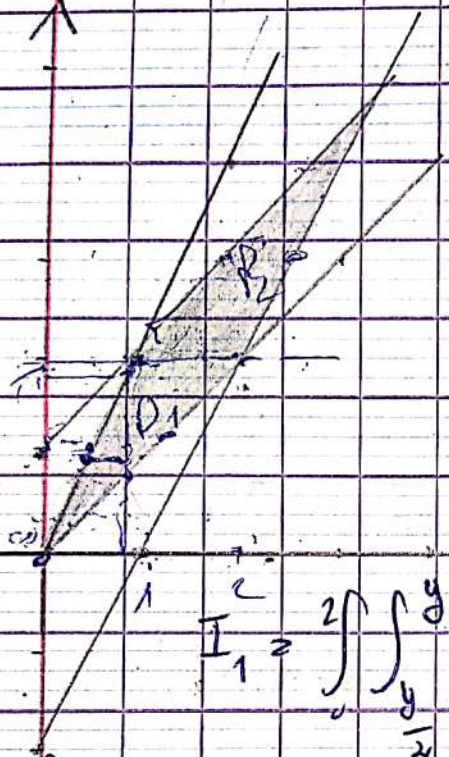
Exercice 7 Calculer le volume V du domaine intérieur à la surface d'équation :

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1 \text{ avec } (a,b,c)\in(\mathbb{R}_+^*)^3.$$

Fiche TD n°1 2021-2022

Exercice ①: 1) $D_1 = \left\{ \frac{y}{2} \leq x \leq y \; ; \; 0 \leq y \leq 2 \right\}$

$D_2 = \left\{ y-1 \leq x \leq \frac{y+2}{2} \; ; \; 2 \leq y \leq 4 \right\}$



$$I = \iint_{D_1} (2x-y)^2 \, dx\,dy + \iint_{D_2} (2x-y)^2 \, dx\,dy$$

$$I = I_1 + I_2$$

$$I_1 = \int_0^2 \int_{\frac{y}{2}}^{y} (2x-y)^2 \, dx\,dy$$

$$I_1 = \int_0^2 \frac{1}{6} (2x-y)^3 \Big|_{\frac{y}{2}}^{y} dy = \frac{1}{6} \int_0^2 y^3 \, dy$$

$$I_1 = \frac{1}{24} y^4 \Big|_0^2 = \frac{16}{24} \Rightarrow \boxed{I_1 = \frac{2}{3}}$$

$$I_2 = \int_2^4 \int_{y-1}^{\frac{y+2}{2}} (2x-y)^2 \, dx\,dy = \int_2^4 \frac{1}{6} (2x-y)^3 \Big|_{y-1}^{\frac{y+2}{2}} dy$$

$$I_2 = \frac{1}{6} \int_2^4 (y+2-y)^3 - (2y-2-y)^3 \, dy$$

$$I_2 = \frac{1}{6} \int_2^4 2^3 \, dy - \frac{1}{6} \int_2^4 (y-2)^3 \, dy$$

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$$I_2 = \frac{8}{6}(4-2) - \frac{1}{24}(y-2)^4 \Big|_2^4$$

$$I_2 = \frac{8}{6}(2) - \frac{1}{24}\left[16 - 0\right]$$

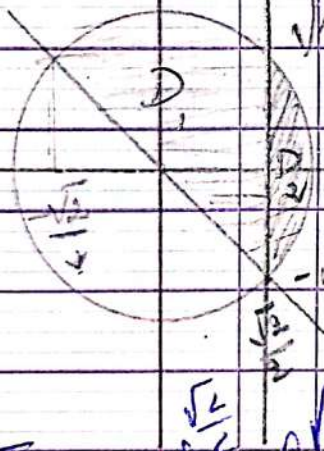
$$I_2 = \frac{8}{3} - \frac{2}{3} = \frac{6}{3} \Rightarrow \boxed{I_2 = \frac{6}{3}}$$

A la fin on déduira que :

$$I = \frac{2}{3} + \frac{6}{3} \Rightarrow \boxed{I = \frac{8}{3}}$$

e) $\iint_D xy^2\,dx\,dy$, $D = \{(x,y)\in\mathbb{R}^2 ;\ x+y \geq 0 \ ;\ x^2+y^2 \leq 1\}$

$D = \{D_1 \cap D_2 \neq \emptyset \ ,\ D_1 \cup D_2\}$



dans : $y \geq -x$ $\quad x^2+y^2 \leq 1$

$\Rightarrow x \leq \left|\frac{\sqrt{2}}{2}\right| \Rightarrow -\frac{\sqrt{2}}{2} \leq x \leq \frac{\sqrt{2}}{2}$

$I = I_1 + I_2$

$$I_1 = \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \int_{-x}^{\sqrt{1-x^2}} xy^2\,dx\,dy = \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \frac{xy^3}{3}\Big|_{-x}^{\sqrt{1-x^2}}\,dx$$

$$I_1 = \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \frac{x}{3}(1-x^2)^{\frac{3}{2}}\,dx + \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \frac{x^4}{3}\,dx$$

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$$I_1 = \frac{-(1+x^2)^{5/2}}{15}\Big|_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} + \frac{x^5}{15}\Big|_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}}$$

$$I_1 = \frac{-\left(\frac{1}{2}\right)^{5/2}}{15} + \frac{\left(\frac{1}{2}\right)^{5/2}}{15} + \frac{1}{15}\left(2\frac{(\sqrt{2})^5}{2^5}\right) = \frac{1}{15}\frac{4\sqrt{2}}{15}$$

$$\boxed{I_1 = \frac{\sqrt{2}}{60}}$$

$$I_2 = \int_{\frac{\sqrt{2}}{2}}^{1} x\left(\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} y^2\,dy\right)dx$$

$$I_2 = \int_{\frac{\sqrt{2}}{2}}^{1} \frac{x}{3}\left(y^3\right)\Big|_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dx = \frac{2}{3}\int_{\frac{\sqrt{2}}{2}}^{1} x(1-x^2)^{\frac{3}{2}}\,dx$$

$$I_2 = \frac{2}{3}\left(-\frac{2}{5}(1-x^2)^{5/2}\right)\Big|_{\frac{\sqrt{2}}{2}}^{1} = -\frac{2}{15}\left(-\frac{1}{2}\right)^{5/2}$$

$$I_2 = \frac{2}{15}\left(\frac{\sqrt{2}}{2}\right)^5 = \frac{\sqrt{2}}{60}$$

$$I = I_1 + I_2$$

$$I = \frac{\sqrt{2}}{60} + \frac{\sqrt{2}}{60} = \frac{\sqrt{2}}{30}$$

$$\boxed{I = \frac{\sqrt{2}}{30}}$$

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4) $\iint_D (x+2y)^2 \, dx\,dy$; $D = \{A(0,0) ; B(1,1) ; C(2,-1)\}$

$D = \{D_1 \cup D_2$ avec $D_1 \cap D_2 = \emptyset\}$

$D_1 : \{0 < x < 1 \quad -\frac{1}{2}x < y < x\}$

$D_2 : \{1 < x < 2 \quad -\frac{1}{2}x < y < 3 - x\}$



$I = I_1 + I_2$

$$I_1 = \int_0^1 \int_{-\frac{1}{2}x}^{x} (x+2y)^2 \, dx = \int_0^1 \frac{1}{6} (x+2y)^3 \Big|_{-\frac{1}{2}x}^{x} dx$$

$$I_1 = \int_0^1 \frac{1}{6}(x+2x)^3 - \frac{1}{6}\left(x+2\left(-\frac{1}{2}x\right)\right)^3 dx$$

$$I_1 = \int_0^1 \frac{24}{6} x^3 \, dx \Rightarrow I_1 = \frac{27}{24} x^4 \Big|_0^1$$

$$I_1 = \frac{27}{24} \Rightarrow \boxed{I_1 = \frac{9}{8}}$$

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$$I_2 = \int_1^2 \int_{-\frac{1}{2}x}^{3-2x} (x+2y)^3 \, dx = \int_1^2 \frac{24}{6} (2-x)^3 \, dx$$

$$I_2 = -\frac{24}{24} (2-x)^4 \Big|_1^2$$

$$I_2 = -\frac{24}{24} (2-2)^4 + \frac{24}{24} (2-1)^4 = \frac{24}{24} = \frac{9}{8}$$

$$\boxed{I_2 = \frac{9}{8}}$$

$$I = I_1 + I_2$$

$$I = \int_0^1 \int_{-\frac{1}{2}x}^{x} (x+2y)^2 \, dx\,dy + \int_1^2 \int_{-\frac{1}{2}x}^{3-2x} (x+2y)^2 \, dx\,dy$$

$$I = \frac{9}{8} + \frac{9}{8} = \frac{18}{8} = \frac{9}{4}$$

$$\boxed{I = \frac{9}{4}}$$

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Exercice ④ :

a/ $D = \{ x^2 + y^2 \leq 1 \ ; \ 0 \leq y \leq x \}$

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ 0 \leq r \leq 1 \end{cases} \quad ; |J_\varphi| = r \quad ; \ 0 \leq \theta \leq \frac{\pi}{4}$$

$$I = \int_0^1 \int_0^{\frac{\pi}{4}} r^3 (\cos\theta - \sin\theta)^2 \, dr \, d\theta$$

$$I = \frac{1}{4} \left[\int_0^{\frac{\pi}{4}} (1 - 2\cos\theta\sin\theta) \, d\theta \right]$$

$$I = \frac{1}{4} \left[\theta \Big|_0^{\frac{\pi}{4}} - \sin^2\theta \Big|_0^{\frac{\pi}{4}} \right] = \frac{1}{4} \left[\frac{\pi}{4} - \frac{1}{2} \right]$$

$$\boxed{I = \frac{\pi - 2}{16}}$$

b/ $$f(x,y) = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right)^{-\frac{1}{2}}$$

$$D = \left\{ \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \ ; \begin{matrix} x = a\rho\cos\theta \\ y = b\rho\sin\theta \end{matrix} \right\}$$

$$I = \int_0^1 \int_0^{2\pi} (\rho^2)^{-\frac{1}{2}} ab\rho \, d\rho \, d\theta$$

$$I = \int_0^1 \int_0^{2\pi} ab \, d\rho \, d\theta \qquad \boxed{= 2ab\pi}$$

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a) $f(x,y) = 1$ $\Rightarrow \begin{cases} y = ax \\ y = bx \end{cases}$, $\begin{cases} y = x/a \\ y = \frac{1}{bx} \end{cases}$

- φ est bijective
- φ est de classe C^1.
- φ^{-1} est de classe C^1.

$x = \frac{u}{v}$, $\frac{x}{a} \leq y \leq ax$

$y = u \cdot v$, $\frac{1}{bx} \leq y \leq \frac{b}{x}$

$\frac{x}{y} = \frac{1}{v^2}$, $xy = u^2$

$\Rightarrow \boxed{\frac{y}{x} = v^2}$ et $\boxed{xy = u^2}$

$\frac{1}{a} \leq \frac{y}{x} \leq a \Rightarrow \frac{1}{a} \leq v^2 \leq a$

$\Rightarrow \frac{1}{\sqrt{a}} \leq v \leq \sqrt{a}$ $\quad |J\varphi| = 2\frac{u}{v}$

$\frac{1}{b} \leq yx \leq b \Rightarrow \frac{1}{b} \leq u^2 \leq b \Rightarrow \frac{1}{\sqrt{b}} \leq u \leq \sqrt{b}$

$\Delta = \left\{ (u,v) \in \mathbb{R}^2 ; \frac{1}{\sqrt{b}} \leq u \leq b ; \frac{1}{\sqrt{a}} \leq v \leq \sqrt{a} \right\}$

$$I = \int_{\frac{1}{\sqrt{b}}}^{\sqrt{b}} \int_{\frac{1}{\sqrt{a}}}^{\sqrt{a}} |J\varphi| \, du \, dv = \int_{\frac{1}{\sqrt{b}}}^{\sqrt{b}} \int_{\frac{1}{\sqrt{a}}}^{\sqrt{a}} 2\frac{u}{v} \, du \, dv$$

$$I = 2\left(\int_{\frac{1}{\sqrt{a}}}^{\sqrt{a}} \frac{1}{v} dv \right) \cdot \left(\int_{\frac{1}{\sqrt{b}}}^{\sqrt{b}} u \, du \right) = \left(b - \frac{1}{b} \right) \left(\frac{1}{2}\ln a + \frac{1}{2}\ln a \right)$$

$$\boxed{I = \left(b - \frac{1}{b} \right) \ln a}$$

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Exercice (3) :

Aire $(D) = \mathcal{A}$

$\iint x \, dx dy = \frac{35}{2}$

$\iint y \, dx dy = \frac{35}{2}$

on sait que :

$x_G = \frac{1}{A} \iint x \, dx dy$

$y_G = \frac{1}{A} \iint y \, dx dy$

$G = \left(\frac{5}{2} ; \frac{5}{2}\right)$

Exercice (4) :

Calculons les coordonnées de centre de Gravité :

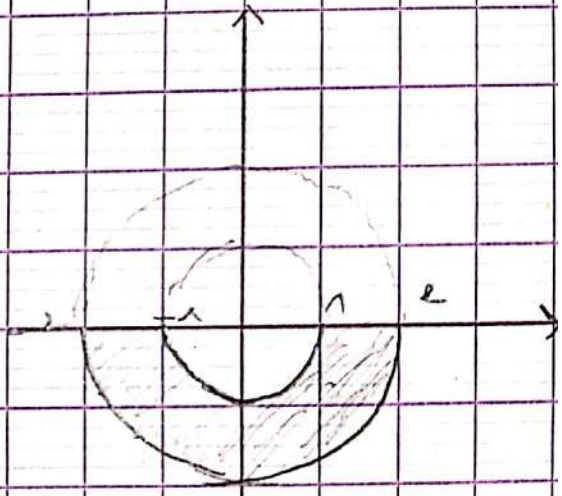
1/ calculons Aire (D) :

$\text{Aire}(D) = \iint_D dx dy$

passons $\begin{cases} x = 2\cos\theta \\ y = 2\sin\theta \end{cases}$ $|J_\varphi| = r$

$1 \leq r \leq 2$

$-\pi \leq \theta \leq 0$



$\text{Aire}(D) = \int_1^2 \int_{-\pi}^{0} r \, dr \, d\theta = \frac{3\pi}{2}$

On a : $x_G = 0$ (symétrie) ; $y_G = \frac{1}{A} \iint_D y \, dx dy$

$y_G = \frac{2}{3\pi} \int_1^2 \int_{-\pi}^{0} r^2 \sin\theta \, dr \, d\theta = \frac{2}{3\pi} \left(\frac{1}{3} r^3 \Big|_1^2\right) \cdot \left(-\cos\theta \Big|_{-\pi}^{0}\right)$

$y_G = \frac{2}{3\pi} \left(\frac{7}{3}\right) (-2)$

$\boxed{= -\frac{28}{9\pi}}$

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Exercice (3) :

Aire $(D) = \frac{\pi}{2}$

$\iint x \, dx dy = \frac{35}{2}$

$\iint y \, dx dy = \frac{35}{2}$

on sait que :

$x_G = \frac{1}{A} \iint x \, dx dy$

$y_G = \frac{1}{A} \iint y \, dx dy$

$G = \left(\frac{5}{2} ; \frac{5}{2}\right)$

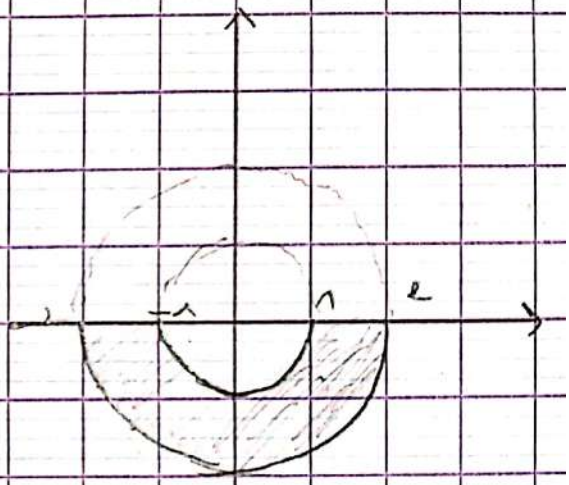
Exercice (4) :

Calculons les coordonnées de centre de Gravité :

1/ calculons Aire (D) :

$\text{Aire}(D) = \iint_D dx dy$

passons $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ 1 \leq r \leq 2 \\ -\pi \leq \theta \leq 0 \end{cases}$ $\quad |J_\varphi| = r$



$\text{Aire}(D) = \int_1^2 \int_{-\pi}^{0} r \, dr \, d\theta = \frac{3\pi}{2}$

On a : $x_G = 0$ (symétrie) ; $y_G = \frac{1}{A} \iint_D y \, dx dy$

$y_G = \frac{2}{3\pi} \int_1^2 \int_{-\pi}^{0} r^2 \sin\theta \, dr \, d\theta = \frac{2}{3\pi} \left(\frac{1}{3} r^3 \Big|_1^2\right) \cdot \left(-\cos\theta \Big|_{-\pi}^{0}\right)$

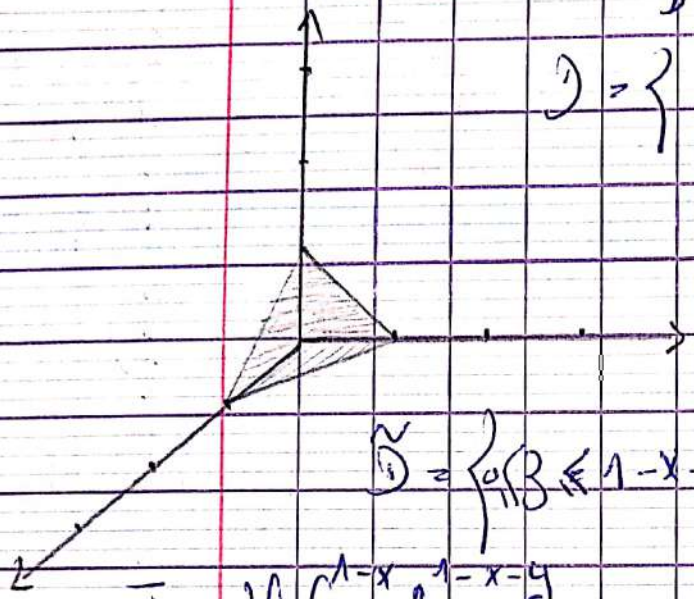
$y_G = \frac{2}{3\pi} \left(\frac{7}{3}\right) (-2)$

$\boxed{= -\frac{28}{9\pi}}$

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Exercice 5 : $I = \iiint_D (x+y+z)^2 \, dx\,dy\,dz$

$D = \{ x=0, y>0, z>0 ; x+y+z=1 \}$



$\tilde{D} = \{ 0 \leq z \leq 1-x-y ; 0 < y < 1-x ; 0 < x < 1 \}$

$$I = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} (x+y+z)^2 \, dx\,dy\,dz$$

$$I = \int_0^1 \int_0^{1-x} \frac{1}{3} (x+y+z)^3 \Big|_0^{1-x-y} dy\,dx$$

$$I = \frac{1}{3} \int_0^1 \int_0^{1-x} \left[(x+y+1-x-y)^3 - (x+y)^3 \right] dy\,dx$$

$$I = \frac{1}{3} \int_0^1 \left(\int_0^{1-x} dy \right) dx - \frac{1}{3} \int_0^1 \left(\int_0^{1-x} (x+y)^3 \, dy \right) dx$$

$$I = \frac{1}{3} x \Big|_0^1 - \frac{1}{6} x^2 \Big|_0^1 - \frac{1}{12} \int_0^1 dx + \frac{1}{12} \int_0^1 x^4 \, dx$$

$$I = \frac{1}{3} - \frac{1}{6} - \frac{1}{12} + \frac{1}{60}$$

$$\boxed{I = \frac{1}{10}}$$

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Exer :

b/ $I = \iiint_D (x-y+z)^2 \, dx\,dy\,dz$

D : limité par les plans $\{ x=0,\ y=0,\ z=0$
$x+z=1,\ y+z=1 \}$

On : $\begin{cases} y+z=1 \\ x+z=1 \end{cases} \Rightarrow x+z = y+z \Rightarrow y=x$

On a : $D = D_1 \cup D_2$

où $D_1 = \{ (x,y,z) \in \mathbb{R}^3 ;\ (x,y) \in \Delta_1 \}$
$0 \leq z \leq 1-x$

$\Delta_1 = \{ 0 \leq x \leq 1 \ ,\ 0 \leq y \leq x \}$

$D_2 = \{ (x,y,z) \in \mathbb{R}^3 \ ,\ (x,y) \in \Delta_2 \}$
$0 \leq z \leq 1-y$

$\Delta_2 = \{ 0 \leq y \leq 1 \quad 0 \leq x \leq y \}$

D_2 est l'image de D_1 par le changement de variables $\Delta_1 \xrightarrow{\varphi} \Delta_2$

$(x,y) \longmapsto (y,x)$

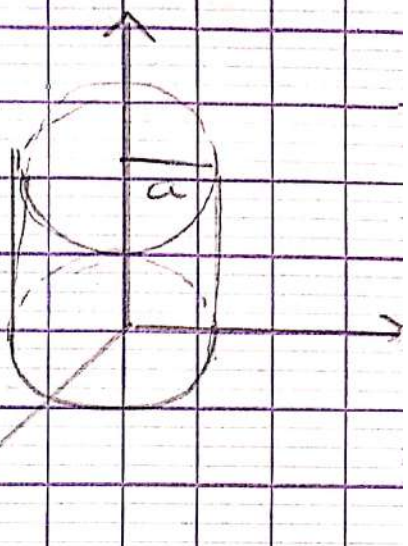
$J_\varphi = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow |J_\varphi| = 1$

$$\boxed{I = \frac{1}{\lambda}}$$

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c/ $\iiint_D \frac{z}{\sqrt{x^2+y^2}}\,dx\,dy\,dz$

$D = \{x^2+y^2 < a^2 \quad 0 < z < a\}$



Coordonné cylindrique :

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases} \qquad |J_\varphi| = r$$

$$I = \int_0^a \int_0^a \int_0^{2\pi} \frac{z}{r}\, r\, dr\, d\theta\, dz = \int_0^a \int_0^{2\pi} \int_0^a z\, dr\, d\theta\, dz$$

$$\boxed{I = a^3\pi}$$

Exercice ② : $I = \iiint_D \frac{z}{\sqrt{x^2+y^2}}\,dx\,dy\,dz$

$D = \{0 < z < b \;;\; 0 < x^2+y^2 < 2ax \;;\; a > 0\}$

$x^2+y^2-2ax < 0 \Rightarrow (x-a)^2 + y^2 < a^2$

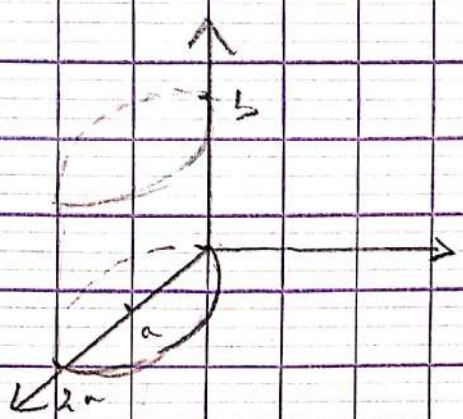
Coordonnées cylindrique :

$x = r\cos\theta$

$y = r\sin\theta$

$z = z$

$|J_\varphi| = r$

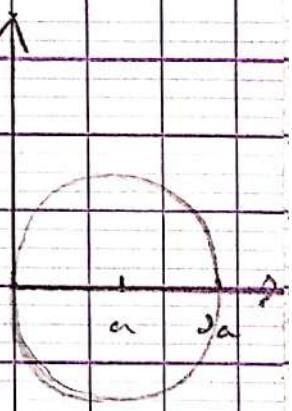


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$x^2 + y^2 < 2ax \qquad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$\Leftrightarrow r^2 < 2ar\cos\theta$

$\Leftrightarrow r < 2a\cos\theta$



$$\Rightarrow \int_0^b \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{2a\cos\theta} \frac{z}{r}\, r\, dr\, d\theta\, dz$$

$$I = \int_0^b z\, dz \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{2a\cos\theta} dr$$

$$I = \frac{b^2}{2} 2a \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\theta\, d\theta = ab^2 \sin\theta \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$\boxed{I = 2ab^2}$$

Exercice (1): Volume intérieur à la surface

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad ; x \geq 0 ; y \geq 0, z \geq 0$$

On pose: $X = \frac{x}{a}$; $Y = \frac{y}{b}$; $Z = \frac{z}{c}$

$$X^2 + Y^2 + Z^2 = 1$$

$$\begin{cases} x = aX \\ y = aY \\ z = aZ \end{cases} \qquad |J_\varphi| = |abc|$$

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$$I = \iiint_D dx\,dy\,dz = \iiint_B |J_\varphi|\,dX\,dY\,dZ$$

$$I = |abc| \iiint_B dX\,dY\,dZ = abc\ vol(B)$$

$$\boxed{I = |abc|\,\frac{4\pi}{3}}$$

Fiche TD : 02 :

Exercice 1 : $F(x^2, y^2, z^2)$

1) $\operatorname{div} F = \vec{\nabla} \cdot \vec{F} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z}$

$\operatorname{div} F = 2(x + y + z)$

$$\operatorname{Rot} F = \vec{\nabla} \wedge \vec{F} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial_x & \partial_y & \partial_z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$= \left(\frac{\partial z^2}{\partial y} - \frac{\partial y^2}{\partial z}\right)\vec{i} + \left(\frac{\partial x^2}{\partial z} - \frac{\partial z^2}{\partial x}\right)\vec{j} + \left(\frac{\partial y^2}{\partial x} - \frac{\partial x^2}{\partial y}\right)\vec{k}$$

$\operatorname{Rot} \vec{F} = \vec{0}$

2) Vu que : $\operatorname{Rot} \vec{F} = \vec{0}$ alors il s'implique que F est un champ de Gradient.

$$\boxed{\vec{F} = \operatorname{grad} f}$$

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